Kriging: Optimal Spatial Prediction

Spatial Data Science

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Lecture Outline (90 minutes)

1. Motivation & Intuition (15 min)
2. The Kriging Framework (25 min)
3. Types of Kriging (15 min)
4. Properties of the Kriging Variance (10 min)
5. Summary & Discussion (5 min)

Why Kriging?

The Spatial Prediction Problem

Given: Sampled values at known locations

Goal: Predict value at an unsampled location

Common alternatives:

- **Nearest neighbor:** Too simple, discontinuous
- **Inverse distance weighting (IDW):** Arbitrary power parameter
- **Spline interpolation:** No uncertainty quantification

Kriging = Optimal solution using the variogram

The Linear Predictor

Kriging Predictor (Ordinary Kriging)

$$\boxed{\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i)}$$

where:

- $\hat{Z}(\mathbf{s}_0)$ = predicted value at unsampled location $\mathbf{s}_0$
- λ_i = kriging weights (to be determined)
- $Z(\mathbf{s}_i)$ = observed values at data locations

Goal: Find weights λ_i that satisfy:

1. **Unbiased:** $\mathbb{E}[\hat{Z}(\mathbf{s}_0) - Z(\mathbf{s}_0)] = 0$
2. **Best:** Minimize prediction variance
3. **Linear:** Linear combination of data

Condition 1: Unbiasedness

We require:

$$\mathbb{E}[\hat{Z}(\mathbf{s}_0) - Z(\mathbf{s}_0)] = 0$$

Substitute the predictor:

$$\mathbb{E}\left[\sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i) - Z(\mathbf{s}_0)\right] = 0$$

Under intrinsic stationarity ($\mathbb{E}[Z(\mathbf{s})] = m$ constant):

$$\sum_{i=1}^{n} \lambda_i m - m = 0$$

Unbiasedness Condition

$$\boxed{\sum_{i=1}^{n} \lambda_i = 1}$$

Prediction variance:

$$\sigma_E^2 = \text{Var}\left(\hat{Z}(\mathbf{s}_0) - Z(\mathbf{s}_0)\right)$$

Using $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$:

$$\sigma_E^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j C_{ij} + C_{00} - 2 \sum_{i=1}^{n} \lambda_i C_{i0}$$

In terms of variogram ($\gamma(h) = C(0) - C(h)$):

$$\sigma_E^2 = 2 \sum_{i=1}^{n} \lambda_i \gamma_{i0} - \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j \gamma_{ij}$$

where $\gamma_{ij} = \gamma(\mathbf{s}_i - \mathbf{s}_j)$

The Kriging System (Ordinary Kriging)

Minimize σ_E^2 subject to $\sum \lambda_i = 1$ using Lagrange multiplier μ:

Ordinary Kriging System

$$\begin{cases} \sum_{j=1}^{n} \lambda_j \gamma(\mathbf{s}_i - \mathbf{s}_j) + \mu = \gamma(\mathbf{s}_i - \mathbf{s}_0), \quad i = 1, \ldots, n \\ \sum_{j=1}^{n} \lambda_j = 1 \end{cases}$$

n equations + 1 constraint = $n + 1$ unknowns $(\lambda_1, \ldots, \lambda_n, \mu)$

Condition 2: Minimum Variance - Setup

Goal

Find weights λ_i that minimize the prediction variance:

$$\sigma_E^2 = \text{Var}\left(\hat{Z}(\mathbf{s}_0) - Z(\mathbf{s}_0)\right)$$

subject to $\sum \lambda_i = 1$ (unbiasedness)

Step 1: Express the prediction error:

$$\hat{Z}(\mathbf{s}_0) - Z(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i) - Z(\mathbf{s}_0)$$

Step 2: Variance of a Linear Combination

Recall from probability theory:

$$\text{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \text{Cov}(X_i, X_j)$$

Apply to our prediction error:

$$\sigma_E^2 = \text{Var}\left(\sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i) - Z(\mathbf{s}_0)\right)$$

Let $Y = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i) - Z(\mathbf{s}_0)$. Then:

$$\sigma_E^2 = \underbrace{\text{Var}\left(\sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i)\right)}_{\text{(A)}} + \underbrace{\text{Var}(Z(\mathbf{s}_0))}_{\text{(B)}} - 2\,\underbrace{\text{Cov}\left(\sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i), Z(\mathbf{s}_0)\right)}_{\text{(C)}}$$

Step 3: Expand Each Term

Term (A): Variance of the weighted sum

$$\mathrm{Var}\left(\sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i)\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j C(\mathbf{s}_i - \mathbf{s}_j)$$

where $C(\mathbf{h})$ is the covariance function.

Term (B): Variance at the prediction location

$$\mathrm{Var}(Z(\mathbf{s}_0)) = C(0) = \text{total variance}$$

Term (C): Covariance between prediction and observation

$$\mathrm{Cov}\left(\sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i), Z(\mathbf{s}_0)\right) = \sum_{i=1}^{n} \lambda_i C(\mathbf{s}_i - \mathbf{s}_0)$$

Step 4: Combine Terms

Substitute (A), (B), and (C) into σ_E^2:

$$\sigma_E^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j C_{ij} + C_{00} - 2 \sum_{i=1}^{n} \lambda_i C_{i0}$$

where $C_{ij} = C(\mathbf{s}_i - \mathbf{s}_j)$.

This is the prediction variance in terms of the **covariance**.

But variogram is often preferred over covariance in geostatistics.

Step 5: Convert from Covariance to Variogram

Recall the fundamental relationship (from earlier lecture):

$$\gamma(\mathbf{h}) = C(0) - C(\mathbf{h})$$

Therefore:

$$C_{ij} = C(0) - \gamma_{ij}, \quad C_{i0} = C(0) - \gamma_{i0}, \quad C_{00} = C(0)$$

Substitute into the variance expression:

$$\sigma_E^2 = \sum_i \sum_j \lambda_i \lambda_j [C(0) - \gamma_{ij}] + C(0) - 2 \sum_i \lambda_i [C(0) - \gamma_{i0}]$$

Expand and use $\sum \lambda_i = 1$:

$$\sigma_E^2 = C(0) \underbrace{\sum_i \sum_j \lambda_i \lambda_j}_{=1} - \sum_i \sum_j \lambda_i \lambda_j \gamma_{ij} + C(0) - 2C(0) \underbrace{\sum_i \lambda_i}_{=1} + 2 \sum_i \lambda_i \gamma_{i0}$$

Simplify term by term:

$$\sigma_E^2 = C(0) - \sum_i \sum_j \lambda_i \lambda_j \gamma_{ij} + C(0) - 2C(0) + 2 \sum_i \lambda_i \gamma_{i0}$$

$$\sigma_E^2 = 2 \sum_i \lambda_i \gamma_{i0} - \sum_i \sum_j \lambda_i \lambda_j \gamma_{ij}$$

Step 7: Final Expression for Prediction Variance

Prediction Variance in Terms of the Variogram

$$\boxed{\sigma_E^2 = 2\sum_{i=1}^{n} \lambda_i \gamma(\mathbf{s}_i - \mathbf{s}_0) - \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j \gamma(\mathbf{s}_i - \mathbf{s}_j)}$$

Interpretation:

- First term: average variogram between data and prediction point
- Second term: average variogram among data points (redundancy penalty)
- $\sum \lambda_i = 1$ ensures unbiasedness

We want to minimize σ_E^2 subject to $\sum \lambda_i = 1$.

Method: Lagrange multipliers

$$L(\lambda_1, \ldots, \lambda_n, \mu) = \sigma_E^2 - 2\mu \left(\sum_{i=1}^{n} \lambda_i - 1 \right)$$

The factor 2μ is for convenience (simplifies derivatives).

Take partial derivatives:

$$\frac{\partial L}{\partial \lambda_i} = 0 \quad \text{for } i = 1, \ldots, n$$

$$\frac{\partial L}{\partial \mu} = 0 \quad \Rightarrow \quad \sum \lambda_i = 1$$

Step 9: The Derivatives

Compute $\frac{\partial \sigma_E^2}{\partial \lambda_i}$:

$$\frac{\partial}{\partial \lambda_i}\left[2\sum_{k=1}^{n}\lambda_k\gamma_{k0}\right] = 2\gamma_{i0}$$

$$\frac{\partial}{\partial \lambda_i}\left[\sum_{k=1}^{n}\sum_{l=1}^{n}\lambda_k\lambda_l\gamma_{kl}\right] = 2\sum_{j=1}^{n}\lambda_j\gamma_{ij}$$

Therefore:

$$\frac{\partial \sigma_E^2}{\partial \lambda_i} = 2\gamma_{i0} - 2\sum_{j=1}^{n}\lambda_j\gamma_{ij}$$

Step 10: Setting Derivatives to Zero

Set $\frac{\partial L}{\partial \lambda_i} = 0$:

$$\frac{\partial \sigma_E^2}{\partial \lambda_i} - 2\mu = 0$$

$$\left(2\gamma_{i0} - 2\sum_{j=1}^{n} \lambda_j \gamma_{ij}\right) - 2\mu = 0$$

Divide by 2:

$$\gamma_{i0} - \sum_{j=1}^{n} \lambda_j \gamma_{ij} - \mu = 0$$

Rearrange:

$$\sum_{j=1}^{n} \lambda_j \gamma_{ij} + \mu = \gamma_{i0}, \quad i = 1, \ldots, n$$

Step 11: The Kriging System

Combined with the unbiasedness constraint:

Ordinary Kriging System

$$\begin{cases} \sum_{j=1}^{n} \lambda_j \gamma(\mathbf{s}_i - \mathbf{s}_j) + \mu = \gamma(\mathbf{s}_i - \mathbf{s}_0), \quad i = 1, \ldots, n \\ \sum_{j=1}^{n} \lambda_j = 1 \end{cases}$$

This is a system of $n+1$ linear equations with $n+1$ unknowns: - $\lambda_1, \ldots, \lambda_n$ (kriging weights) - μ (Lagrange multiplier)

Step 12: The Kriging Variance (Revisited)

Once we solve for the weights, the minimized prediction variance is:

$$\sigma_K^2 = \sigma_E^2 \Big|_{\text{optimal } \lambda_i}$$

Substitute the kriging system into the variance expression:

$$\sigma_K^2 = \sum_{i=1}^{n} \lambda_i \gamma(\mathbf{s}_i - \mathbf{s}_0) + \mu$$

This is the **kriging variance** a measure of prediction uncertainty.

Summary: Derivation Flow

Start with prediction error: $\hat{Z} - Z$
Take variance: $\sigma_E^2 = \text{Var}(\hat{Z} - Z)$
Express in terms of covariance
Convert to variogram: $\gamma(h) = C(0) - C(h)$
Apply unbiasedness: $\sum \lambda_i = 1$
Simplify to: $\sigma_E^2 = 2\sum \lambda_i \gamma_{i0} - \sum_i \sum_j \lambda_i \lambda_j \gamma_{ij}$
Minimize using Lagrange multipliers
Obtain Kriging System and Kriging Variance

Matrix Form of the Kriging System

$$\begin{bmatrix} \gamma_{11} & \gamma_{12} & \cdots & \gamma_{1n} & 1 \\ \gamma_{21} & \gamma_{22} & \cdots & \gamma_{2n} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \gamma_{n1} & \gamma_{n2} & \cdots & \gamma_{nn} & 1 \\ 1 & 1 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \\ \mu \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \gamma_{20} \\ \vdots \\ \gamma_{n0} \\ 1 \end{bmatrix}$$

$$\boxed{\begin{array}{c} \mathbf{A}\boldsymbol{\lambda} = \mathbf{b} \\ \text{Solve for } \boldsymbol{\lambda} = \mathbf{A}^{-1}\mathbf{b} \end{array}}$$

The Kriging Variance

Once weights are known, the minimized prediction variance is:

Kriging Variance

$$\boxed{\sigma_K^2(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i \gamma(\mathbf{s}_i - \mathbf{s}_0) + \mu}$$

Important properties:

- Does **not** depend on data values
- Only depends on:
  - Variogram model (nugget, sill, range)
  - Configuration of data points
  - Location of prediction point
- $\sigma_K^2(\mathbf{s}_i) = 0$ at data locations (exact interpolation)

What Does the Kriging Variance Tell Us?

$$\boxed{\sigma_K^2(\mathbf{s}_0) = \text{measure of estimation reliability}}$$

Properties:

1. **Zero at data locations** (exact interpolator)

$$\sigma_K^2(\mathbf{s}_i) = 0$$

2. **Increases with distance from data**

$$\text{Far from samples} \Rightarrow \text{Higher uncertainty}$$

3. **Larger in sparse data areas**
4. **Depends on variogram parameters**

$$\text{Higher nugget} \Rightarrow \text{Higher variance}$$

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